Year 10 Foundation

REVISION PACK

Progress Check 3

Thursday 19 November 2026

Name: _____

Class: _____

Units in this pack

- Number 3
- Algebra 2
- Graphs 2

Number 3 · Laws of indices (Lesson 6.1)

Worked example: writing as a single power

Write as a single power of the number.

(a) $6^3 \times 6^4$ (b) $9^8 \div 9^2$ (c) $(2^3)^4$

(a)	**Step 1**	Multiply: add the powers	$3 + 4 = 7$
	Step 2	Answer	6^7
(b)	**Step 1**	Divide: subtract the powers	$8 - 2 = 6$
	Step 2	Answer	9^6
(c)	**Step 1**	Power of a power: multiply	$3 \times 4 = 12$
	Step 2	Answer	2^{12}

Practice: writing as a single power

1.1 Write as a single power of the number.

(a) $4^5 \times 4^2$ (b) $8^{10} \div 8^3$ (c) $(5^2)^3$

1.2 Write as a single power of the number.

(a) $6^2 \times 6^7$ (b) $9^8 \div 9^5$ (c) $(7^4)^2$

1.3 Write as a single power of the number.

(a) $3^4 \times 3^6$ (b) $2^9 \div 2^2$ (c) $(2^5)^3$

1.4 Write as a single power of the number.

(a) $10^3 \times 10^5$ (b) $5^{12} \div 5^4$ (c) $(4^3)^4$

1.5 Write as a single power of the number.

(a) $7^2 \times 7^8$ (b) $3^{11} \div 3^6$ (c) $(6^2)^6$

1.6 Write as a single power of the number.

(a) $2^6 \times 2^5$ (b) $6^9 \div 6^3$ (c) $(9^3)^3$

Number 3 · Laws of indices (Lesson 6.1)

Worked example: negative and zero powers

Work out the value of

(a) 3^{-2} (b) 8^0

(a)	**Step 1**	Minus power: 1 over	$3^{-2} = \frac{1}{3^2}$
	Step 2	Answer	$\frac{1}{9}$
(b)	**Step 1**	Power 0 is always 1	$8^0 = 1$
	Step 2	Answer	1

Practice: negative and zero powers

1.7 Work out the value of

(a) 4^{-2} (b) 7^0

1.8 Work out the value of

(a) 2^{-4} (b) 25^0

1.9 Work out the value of

(a) 10^{-2} (b) 9^0

1.10 Work out the value of

(a) 3^{-3} (b) 15^0

1.11 Work out the value of

(a) 5^{-2} (b) 100^0

1.12 Work out the value of

(a) 6^{-1} (b) 48^0

Worked example: simplifying with index laws

Simplify

(a) $x^4 \times x^6$ (b) $y^9 \div y^3$ (c) $(3a^4)^2$ (d) $\dfrac{10a^6b^3}{2a^2b}$

(a)	**Step 1**	Add the powers	$4 + 6 = 10$
	Step 2	Answer	x^{10}
(b)	**Step 1**	Subtract the powers	$9 - 3 = 6$
	Step 2	Answer	y^6
(c)	**Step 1**	Number	$3^2 = 9$
	Step 2	Letter: multiply powers	$4 \times 2 = 8$
	Step 3	Answer	$9a^8$
(d)	**Step 1**	Numbers	$10 \div 2 = 5$
	Step 2	Letters: subtract powers	$a^{6-2} = a^4, \quad b^{3-1} = b^2$
	Step 3	Answer	$5a^4b^2$

Practice: simplifying with index laws

2.1 Simplify

(a) $x^2 \times x^7$ (b) $y^{10} \div y^4$ (c) $(5x^2)^2$ (d) $\dfrac{8m^7n^3}{2m^3n}$

2.2 Simplify

(a) $a^5 \times a^3$ (b) $b^7 \div b^2$ (c) $(2a^4)^3$ (d) $\dfrac{15x^5y^4}{3x^2y}$

2.3 Simplify

(a) $p^4 \times p^6$ (b) $q^9 \div q^3$ (c) $(4y^5)^2$ (d) $\dfrac{12p^9q^2}{4p^4q^2}$

2.4 Simplify

(a) $m^3 \times m^8$ (b) $n^{12} \div n^5$ (c) $(3x^2)^3$ (d) $\dfrac{20a^6b^4}{5a^2b^3}$

2.5 Simplify

(a) $x^6 \times x^4$ (b) $y^{11} \div y^2$ (c) $(6b^3)^2$ (d) $\dfrac{18m^5n^3}{9m^4n}$

2.6 Simplify

(a) $t^7 \times t$ (b) $k^{15} \div k^5$ (c) $(2c^5)^4$ (d) $\dfrac{14x^8y^5}{7x^3y^2}$

Graphs 2 · Conversion graphs (Lesson 8.1)

Worked example: using a conversion graph

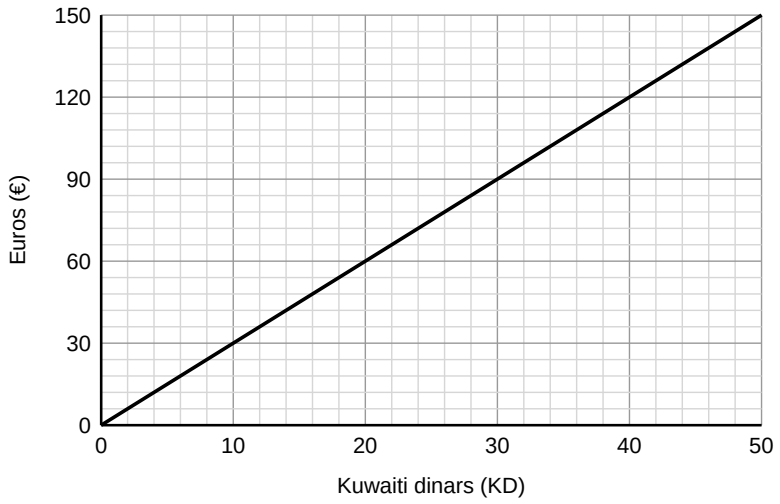
From a conversion graph, 10 KD = €40.

(a) Change 30 KD into euros. (b) Change €60 into KD. (c) Change 500 KD into euros.

(a)	**Step 1**	Up from 30 KD, across	$30\ \text{KD} = €120$
	Step 2	Answer	$€120$
(b)	**Step 1**	Across from €60, down	$€60 = 15\ \text{KD}$
	Step 2	Answer	$15\ \text{KD}$
(c)	**Step 1**	Off the graph: split	$500 = 50 \times 10$
	Step 2	Use 10 KD = €40	$50 \times 40 = 2000$
	Step 3	Answer	$€2000$

Practice: using a conversion graph

3.1



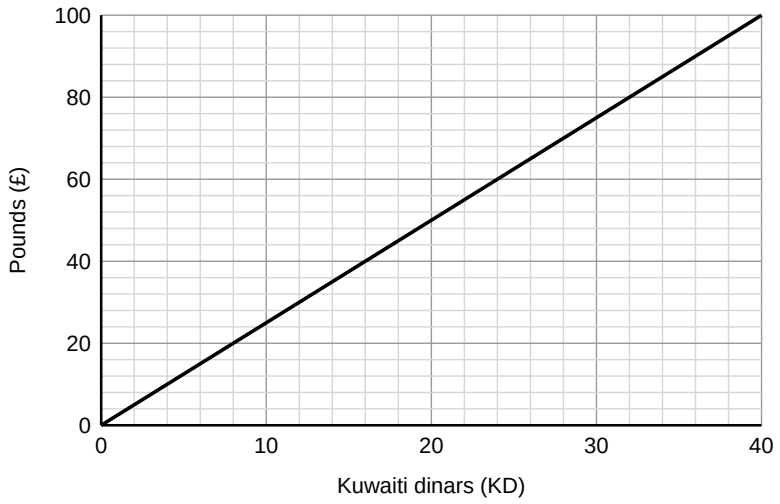
You can use this graph to change between Kuwaiti dinars (KD) and Euros (€).

(a) Change 20 KD into euros.

(b) Change €90 into KD.

(c) Change 300 KD into euros.

3.2



You can use this graph to change between Kuwaiti dinars (KD) and Pounds (£).

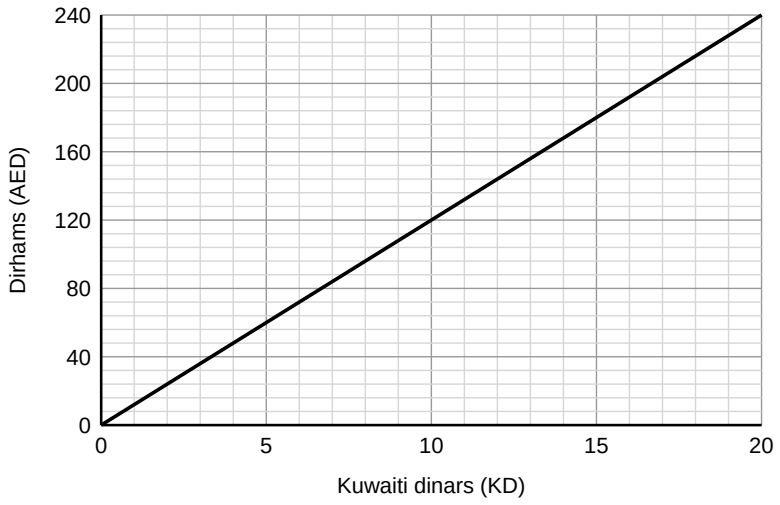
(a) Change 16 KD into pounds.

(b) Change £70 into KD.

(c) Change 400 KD into pounds.

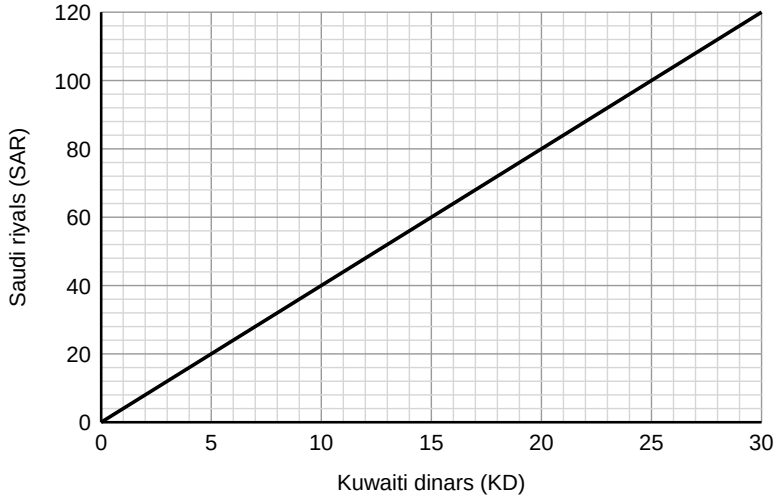
Practice: using a conversion graph (continued)

3.3



You can use this graph to change between Kuwaiti dinars (KD) and Dirhams (AED).
(a) Change 15 KD into dirhams.
(b) Change AED 96 into KD.
(c) Change 200 KD into dirhams.

3.4



You can use this graph to change between Kuwaiti dinars (KD) and Saudi riyals (SAR).
(a) Change 25 KD into saudi riyals.
(b) Change SAR 60 into KD.
(c) Change 150 KD into saudi riyals.

Graphs 2 · Distance-time graphs (Lesson 8.2)

Worked example: reading a distance-time graph

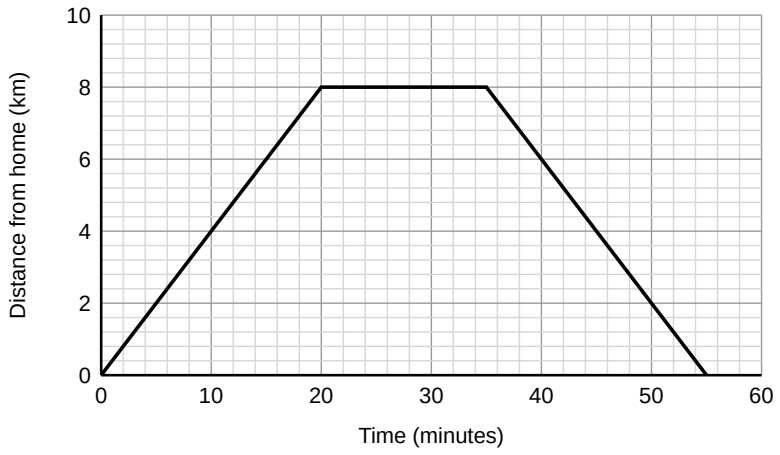
A cyclist rides 6 km from home in 20 minutes, stops until 30 minutes, then rides home in 30 minutes.

(a) How long did they stop? (b) How far from home after 10 minutes? (c) Speed on the way home in km/h?

(a)	Step 1	Flat line = stopped	$30 - 20 = 10$
	Step 2	Answer	10 minutes
(b)	Step 1	Up from 10 min, across	3 km
	Step 2	Answer	3 km
(c)	Step 1	Time in hours	30 min $= \frac{1}{2}$ h
	Step 2	Distance ÷ time	$6 \div \frac{1}{2} = 12$
	Step 3	Answer	12 km/h

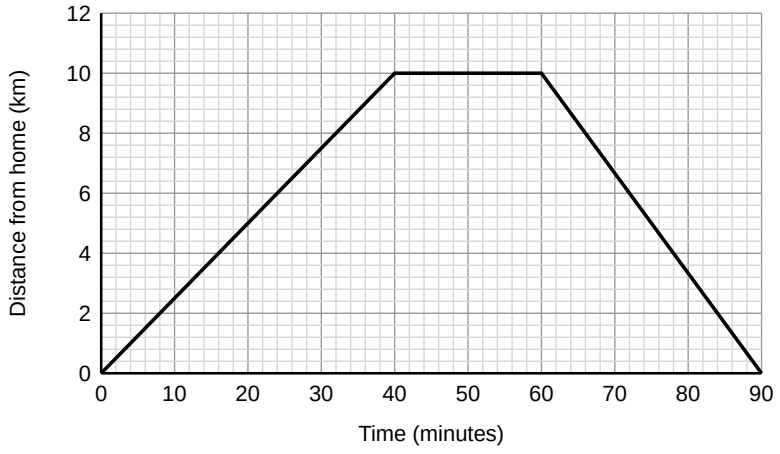
Practice: reading a distance-time graph

4.1



Ali cycles from home to a shop and back.
The distance-time graph shows the journey.
(a) How long did Ali stay at a shop?
(b) How far from home was Ali after 10 minutes?
(c) Work out Ali's average speed on the way home.
Give your answer in km/h.

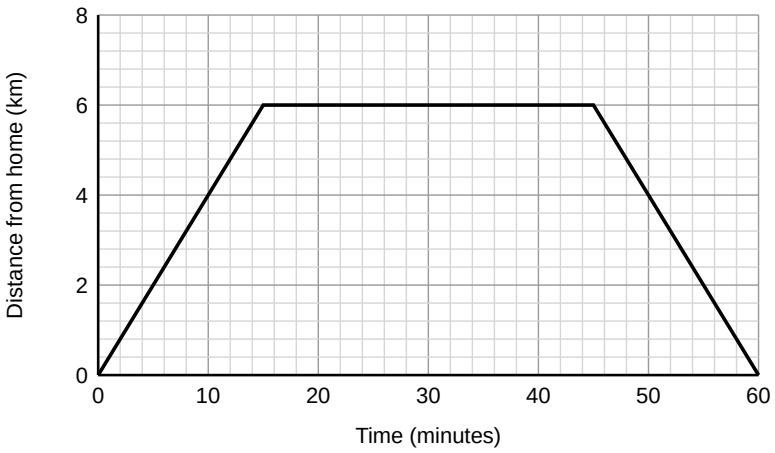
4.2



Sara cycles from home to a friend's house and back.
The distance-time graph shows the journey.
(a) How long did Sara stay at a friend's house?
(b) How far from home was Sara after 20 minutes?
(c) Work out Sara's average speed on the way home.
Give your answer in km/h.

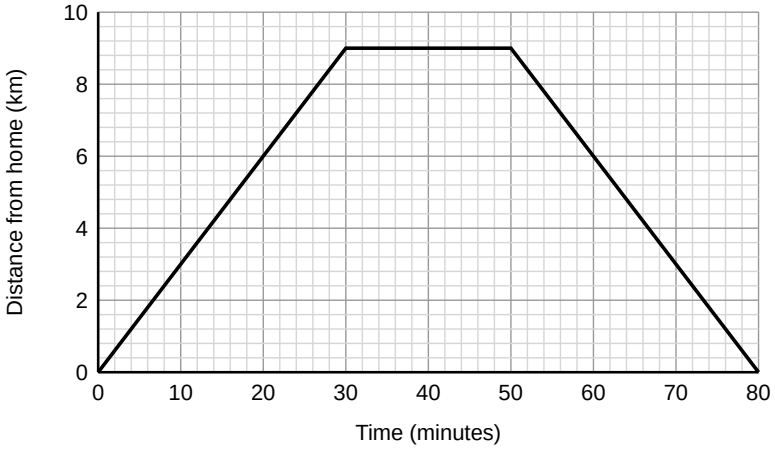
Practice: reading a distance-time graph (continued)

4.3



Omar cycles from home to the beach and back.
The distance-time graph shows the journey.
(a) How long did Omar stay at the beach?
(b) How far from home was Omar after 5 minutes?
(c) Work out Omar's average speed on the way home.
Give your answer in km/h.

4.4



Huda cycles from home to the library and back.
The distance-time graph shows the journey.
(a) How long did Huda stay at the library?
(b) How far from home was Huda after 10 minutes?
(c) Work out Huda's average speed on the way home.
Give your answer in km/h.

Worked example: acceleration and constant speed

A car speeds up from 0 to 15 m/s in 5 s, then travels at 15 m/s for 8 s.

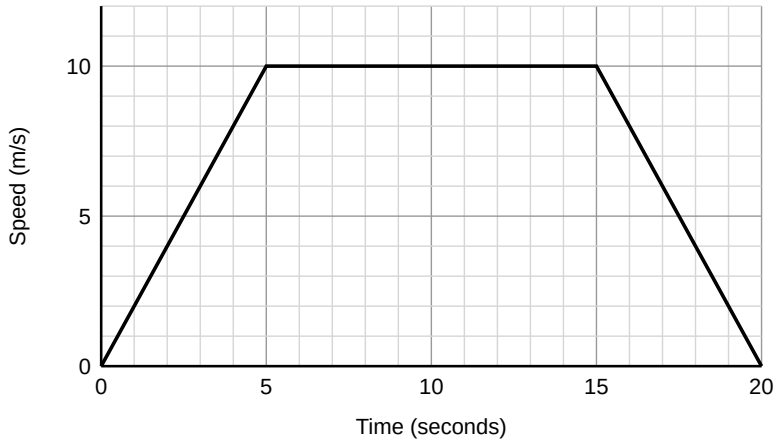
(a) Work out the acceleration in the first 5 seconds.

(b) Work out the distance at constant speed.

(a)	Step 1	Speed ÷ time	$15 \div 5 = 3$
	Step 2	Answer	$3\ \text{m/s}^2$
(b)	Step 1	Speed × time	$15 \times 8 = 120$
	Step 2	Answer	$120\ \text{m}$

Practice: acceleration and constant speed

5.1

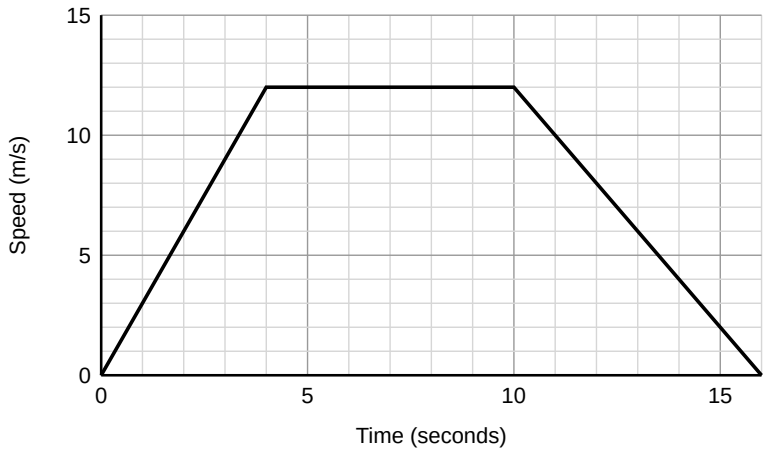


The speed-time graph shows a car journey of 20 seconds.

(a) Work out the acceleration of the car during the first 5 seconds.

(b) Work out the distance the car travels while it is moving at a constant speed.

5.2



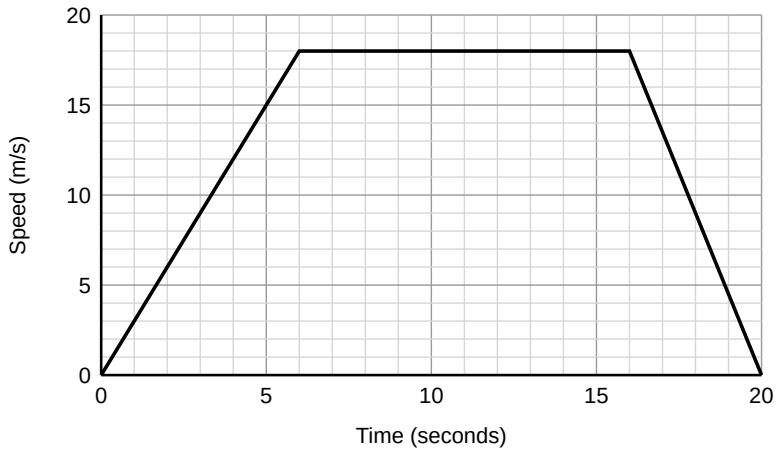
The speed-time graph shows a car journey of 16 seconds.

(a) Work out the acceleration of the car during the first 4 seconds.

(b) Work out the distance the car travels while it is moving at a constant speed.

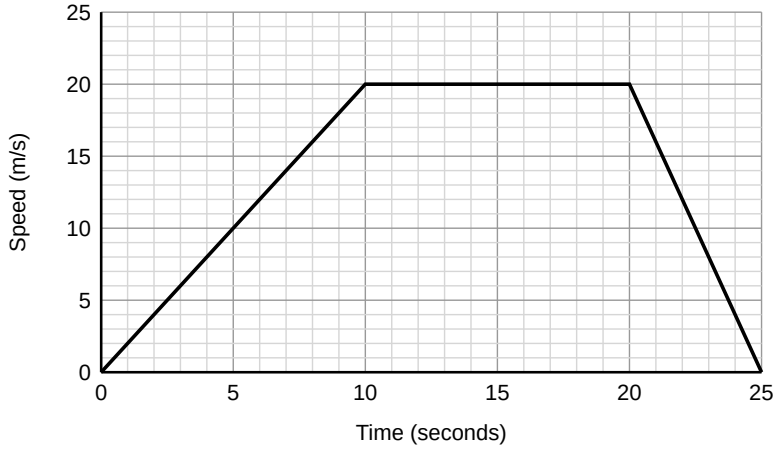
Practice: acceleration and constant speed (continued)

5.3



The speed-time graph shows a car journey of 20 seconds.
(a) Work out the acceleration of the car during the first 6 seconds.
(b) Work out the distance the car travels while it is moving at a constant speed.

5.4



The speed-time graph shows a car journey of 25 seconds.
(a) Work out the acceleration of the car during the first 10 seconds.
(b) Work out the distance the car travels while it is moving at a constant speed.

Graphs 2 · Mid-points and perpendicular lines (Lessons 8.4, 8.5)

Worked example: mid-point of a line

A is the point (4, 5). B is the point (10, 9). Find the coordinates of the midpoint of AB.

Step 1	Add x, halve	$\frac{4+10}{2} = 7$
Step 2	Add y, halve	$\frac{5+9}{2} = 7$
Step 3	Answer	$(7, 7)$

Practice: mid-point of a line

6.1 A is the point (4, 1). B is the point (10, 7). Find the coordinates of the midpoint of AB.

6.2 A is the point (1, 6). B is the point (9, 12). Find the coordinates of the midpoint of AB.

6.3 A is the point (3, 2). B is the point (9, 12). Find the coordinates of the midpoint of AB.

6.4 A is the point (2, 4). B is the point (6, 10). Find the coordinates of the midpoint of AB.

6.5 A is the point (0, 3). B is the point (8, 5). Find the coordinates of the midpoint of AB.

6.6 A is the point (5, 1). B is the point (11, 9). Find the coordinates of the midpoint of AB.

Graphs 2 · Mid-points and perpendicular lines (Lessons 8.4, 8.5)

Worked example: perpendicular lines

This question is about perpendicular lines.

(a) A line has gradient 5. Write down the gradient of a line perpendicular to it.

(b) Show that the lines $y = 2x + 1$ and $y = -\frac{1}{2}x + 3$ are perpendicular.

(a)	Step 1	Flip	$\frac{1}{5}$
	Step 2	Change the sign	$-\frac{1}{5}$
	Step 3	Answer	$-\frac{1}{5}$
(b)	Step 1	Gradients	2 and $-\frac{1}{2}$
	Step 2	Multiply	$2 \times -\frac{1}{2} = -1$
	Step 3	Answer	They multiply to -1, so perpendicular

Practice: perpendicular lines

6.7 (a) A line has gradient 2. Write down the gradient of a line perpendicular to it.

(b) Show that the lines $y = 4x + 1$ and $y = -\frac{1}{4}x + 5$ are perpendicular.

6.8 (a) A line has gradient 6. Write down the gradient of a line perpendicular to it.

(b) Show that the lines $y = 5x - 2$ and $y = -\frac{1}{5}x + 3$ are perpendicular.

6.9 (a) A line has gradient $\frac{1}{3}$. Write down the gradient of a line perpendicular to it.

(b) Show that the lines $y = \frac{1}{2}x + 4$ and $y = -2x + 1$ are perpendicular.

6.10 (a) A line has gradient 3. Write down the gradient of a line perpendicular to it.

(b) Show that the lines $y = 6x + 1$ and $y = -\frac{1}{6}x + 2$ are perpendicular.

6.11 (a) A line has gradient $\frac{1}{4}$. Write down the gradient of a line perpendicular to it.

(b) Show that the lines $y = -5x + 2$ and $y = \frac{1}{5}x + 1$ are perpendicular.

6.12 (a) A line has gradient -2. Write down the gradient of a line perpendicular to it.

(b) Show that the lines $y = \frac{2}{3}x + 1$ and $y = -\frac{3}{2}x + 4$ are perpendicular.

Number 3 · Using index laws (Lesson 6.1)

Worked example: working out with powers

Work out

(a) $5^2 \times 5^4$. Give your answer as a power of 5. (b) the value of $2^5 \div 2$ (c) the value of 7^0

(a)	**Step 1**	Add the powers	$2 + 4 = 6$
	Step 2	Answer	5^6
(b)	**Step 1**	Subtract the powers	$2^{5-1} = 2^4$
	Step 2	Work it out	$2^4 = 16$
	Step 3	Answer	16
(c)	**Step 1**	Power 0 is always 1	$7^0 = 1$
	Step 2	Answer	1

Practice: working out with powers

7.1 Work out

(a) $2^4 \times 2^3$. Give your answer as a power of 2. (b) the value of $5^3 \div 5$ (c) the value of 9^0

7.2 Work out

(a) $4^2 \times 4^5$. Give your answer as a power of 4. (b) the value of $3^4 \div 3$ (c) the value of 12^0

7.3 Work out

(a) $6^3 \times 6^2$. Give your answer as a power of 6. (b) the value of $2^5 \div 2$ (c) the value of 50^0

7.4 Work out

(a) $7^5 \times 7^2$. Give your answer as a power of 7. (b) the value of $10^4 \div 10$ (c) the value of 3^0

7.5 Work out

(a) $5^2 \times 5^6$. Give your answer as a power of 5. (b) the value of $6^3 \div 6$ (c) the value of 8^0

7.6 Work out

(a) $3^5 \times 3^4$. Give your answer as a power of 3. (b) the value of $2^6 \div 2$ (c) the value of 11^0